

Homework Assignment 8

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1. (3A.17) Suppose that (X, S, μ) is a measure space and $f_1, f_2, \dots$ is a sequence of non-negative S -measurable functions on X . Define a function $f : X \rightarrow [0, \infty]$ by $f(x) = \liminf_{k \rightarrow \infty} f_k(x)$.

(a) Show that f is an S -measurable function.

(b) Prove that

$$\int f \, d\mu \leq \liminf_{k \rightarrow \infty} \int f_k \, d\mu.$$

(c) Give an example showing that the inequality in (b) can be a strict inequality even when $\mu(X) < \infty$ and the family of functions $\{f_k\}_{k \in \mathbb{Z}}$ is uniformly bounded.

Proof.

(a) This falls straight out of the definition:

$$\liminf_{k \rightarrow \infty} f_k = \lim_{k \rightarrow \infty} \left(\inf_{n \geq k} f_n \right).$$

Indeed, the infimums are measurable by the result that the infimum of a sequence of measurable functions is measurable, and the limit of them is measurable due to the result that the limit of measurable functions is measurable.

(b) Define $g_k = \inf_{n \geq k} f_n$ for each k . Then $f_k \geq g_k$ for all k , so

$$\int_X f_k \, d\mu \geq \int_X g_k \, d\mu.$$

Also, $\{g_k\}_1^\infty$ is an increasing sequence of nonnegative functions, and thus $\{\int g_k\}_1^\infty$ is

also increasing, so the Monotone Convergence Theorem gives us

$$\begin{aligned}
\liminf_{k \rightarrow \infty} \int f_k \, d\mu &\geq \liminf_{k \rightarrow \infty} \int g_k \, d\mu \\
&= \lim_{k \rightarrow \infty} \int g_k \, d\mu \\
&= \int \lim_{k \rightarrow \infty} g_k \, d\mu \\
&= \int \lim_{k \rightarrow \infty} \inf_{n \geq k} f_n \, d\mu \\
&= \int \liminf_{k \rightarrow \infty} f_k \, d\mu \\
&= \int f \, d\mu.
\end{aligned}$$

(c) This one is eluding me.

□

2. (3A.20) Suppose (X, S, μ) is a measure space and $f_1, f_2, \dots$ is a monotone (meaning either increasing or decreasing) sequence of S -measurable functions. Define $f : X \rightarrow [-\infty, \infty]$ by

$$f(x) = \lim_{k \rightarrow \infty} f_k(x).$$

Prove that if $\int |f_1| \, d\mu < \infty$, then

$$\lim_{k \rightarrow \infty} \int f_k \, d\mu = \int f \, d\mu.$$

Proof. First, assume $\{f_n\}$ is increasing. Define g_k for each $k \in \mathbb{Z}^+$ by

$$g_k(x) = f_k(x) - f_1(x).$$

Then $\{g_n\}_1^\infty$ is an increasing sequence of nonnegative S -measurable functions, so

$$\lim_{k \rightarrow \infty} \int g_k \, d\mu = \int \lim_{k \rightarrow \infty} g_k \, d\mu$$

by the Monotone Convergence Theorem. Then

$$\begin{aligned}
\int f(x) &= \int \lim_{k \rightarrow \infty} f_k(x) \\
&= \int \lim_{k \rightarrow \infty} (g_k(x) + f_1(x)) \\
&= \int \left(f_1(x) + \lim_{k \rightarrow \infty} g_k(x) \right) \\
&= \int f_1(x) + \int \lim_{k \rightarrow \infty} g_k(x) \\
&= \int f_1(x) + \lim_{k \rightarrow \infty} \int g_k(x) \\
&= \lim_{k \rightarrow \infty} \left(\int f_1(x) + \int g_k(x) \right) \\
&= \lim_{k \rightarrow \infty} \int (f_1(x) + g_k(x)) \\
&= \lim_{k \rightarrow \infty} \int f_k(x).
\end{aligned}$$

The proof when $\{f_n\}$ is decreasing is similar, but different enough to work out on its own. This time, define g_k for each $k \in \mathbb{Z}^+$ by

$$g_k(x) = f_1(x) - f_k(x).$$

Then $\{g_n\}_1^\infty$ is again an increasing sequence of nonnegative S -measurable functions, so

$$\lim_{k \rightarrow \infty} \int g_k \, d\mu = \int \lim_{k \rightarrow \infty} g_k \, d\mu.$$

Then

$$\begin{aligned} \int f(x) &= \int \lim_{k \rightarrow \infty} f_k(x) \\ &= \int \lim_{k \rightarrow \infty} (f_1(x) - g_k(x)) \\ &= \int \left(f_1(x) - \lim_{k \rightarrow \infty} g_k(x) \right) \\ &= \int f_1(x) - \int \lim_{k \rightarrow \infty} g_k(x) \\ &= \int f_1(x) - \lim_{k \rightarrow \infty} \int g_k(x) \\ &= \lim_{k \rightarrow \infty} \left(\int f_1(x) - \int g_k(x) \right) \\ &= \lim_{k \rightarrow \infty} \int (f_1(x) - g_k(x)) \\ &= \lim_{k \rightarrow \infty} \int f_k(x). \end{aligned}$$

□

3. (3B.3) Suppose λ is Lebesgue measure on $\mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel measurable function such that $\int |f| \, d\lambda < \infty$. Define $g : \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(x) = \int_{(-\infty, x)} f \, d\lambda.$$

Prove that g is uniformly continuous on $\mathbb{R}$.

Proof. Note that if $b > a$, then

$$\begin{aligned} g(b) - g(a) &= \int_{(-\infty, b)} f \, d\lambda - \int_{(-\infty, a)} f \, d\lambda \\ &= \int_{\mathbb{R}} f \chi_{(-\infty, b)} \, d\lambda - \int_{\mathbb{R}} f \chi_{(-\infty, a)} \, d\lambda \\ &= \int_{\mathbb{R}} f \chi_{(-\infty, b)} - f \chi_{(-\infty, a)} \, d\lambda \\ &= \int_{\mathbb{R}} f \left(\chi_{(-\infty, b)} - \chi_{(-\infty, a)} \right) \, d\lambda \\ &= \int_{\mathbb{R}} f \chi_{[a, b)} \, d\lambda \\ &= \int_{[a, b)} f \, d\lambda. \end{aligned}$$

Also, given $\varepsilon > 0$, there is a $\delta > 0$ such that any $B \in \mathcal{S}$ with $\mu(B) < \delta$ satisfies

$$\int_B g \, d\mu < \varepsilon.$$

Thus whenever a and b satisfy $b - a < \delta$, we get

$$g(b) - g(a) = \int_{[a,b)} f \, d\lambda < \varepsilon$$

as desired. □

4. (3B.5) Let λ denote Lebesgue measure on $\mathbb{R}$. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel measurable function such that $\int |f| \, d\lambda < \infty$. Prove that

$$\lim_{k \rightarrow \infty} \int_{[-k,k]} f \, d\lambda = \int f \, d\lambda.$$

Proof. Define g_k for each $k \in \mathbb{Z}^+$ by $g_k = f\chi_{[-k,k]}$. Then $|g_k(x)| \leq |f(x)|$ for all x , so the sequence g_k is dominated by $|f|$ in the sense of the Dominated Convergence Theorem, and we conclude that

$$\begin{aligned} \lim_{k \rightarrow \infty} \int_{[-k,k]} f \, d\lambda &= \lim_{k \rightarrow \infty} \int f\chi_{[-k,k]} \, d\lambda \\ &= \lim_{k \rightarrow \infty} \int g_k \, d\lambda \\ &= \int \lim_{k \rightarrow \infty} g_k \, d\lambda \\ &= \int f \, d\lambda. \end{aligned}$$

□

5. (3B.11) Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $f \in \mathcal{L}^1(\mu)$. Prove that the set $\{x \in X : f(x) \neq 0\}$ is the countable union of sets with finite μ -measure.

Proof. Consider the sets $E_n = \{x : |f(x)| > 1/n\}$. The fact that $\int |f| \, d\mu < \infty$ forces $\mu(E_n) < \infty$ for all $n \in \mathbb{Z}^+$ since if $\mu(E_n) = \infty$, then

$$\int_{E_n} |f| \, d\mu \geq \int_{E_n} \frac{1}{n} \, d\mu = \infty.$$

Each $x \in X$ for which $f(x) \neq 0$ satisfies $|f(x)| > 1/n$ for some n , and certainly if $|f(x)| > 1/n$ then $f(x) \neq 0$, so we have that

$$\{x \in X : f(x) \neq 0\} = \bigcup_{n=1}^{\infty} E_n$$

as desired. □